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Palaeomagnetic survey of the (Palaeozoic) Shelve inlier and Berwyn Hills, Welsh Borderlands

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Summary. This study completes palaeomagnetic coverage of Lower Palaeozoic igneous rocks within the Welsh Borderlands, and includes data from Ordovician (Llanvirn and Caradoc) volcanics yielding positive fold tests and palaeomagnetic poles in agreement with other British Ordovician studies. A wide range of Late Ordovician–Lower Silurian (Llandovery) intrusions yield directions of magnetization unlike any previously identified from Lower Palaeozoic rocks of Britain, but highly stable to thermal and af demagnetization, and include a reversal of magnetization. The collective results define a large movement relative to the dipole axis over a period of less than 10 Ma between *c.*445 and 425 Ma; part of this movement can be explained in terms of complex rotations of the British region during late Ordovician and early Silurian tectonism within the paratectonic Caledonides.

1 Introduction

Extensive palaeomagnetic studies of British Lower Palaeozoic rocks show that the magnetic palaeofield direction changed from near $D = 180°$ $I = +45°$ to near $D = 205°$, $I = +55°$ between Ordovician and early Silurian times (Briden, Morris & Piper 1973). For much of Ordovician times and from early Silurian to the end of Lower Devonian times the corresponding (south) palaeomagnetic poles lay near 0° E, 10° S and 335° E, 0° S respectively. By accident of initial sampling these studies did not cover a range of igneous intrusions of known or probable Ordovician age which have since yielded quite different directions of magnetization (Thomas & Briden 1976; Piper *et al.* 1978) and shown that the history of apparent polar wander (apw) movements in Lower Palaeozoic times was considerably more complex than originally suspected. Hitherto the interval separating the two mean field directions quoted above from Late Ordovician to early Silurian times has been sparsely covered by palaeomagnetic work and this study includes results from an extensive range of intrusions in the Welsh Borderlands, assigned to this interval. During reconnaissance work the author found that not only did these rocks possess a stable magnetization quite different from that recovered from other British Palaeozoic rocks, but also that there were considerable differences between the separate intrusions; the collective results indicated rapid apparent polar movements. It became clear that thorough sampling of all available

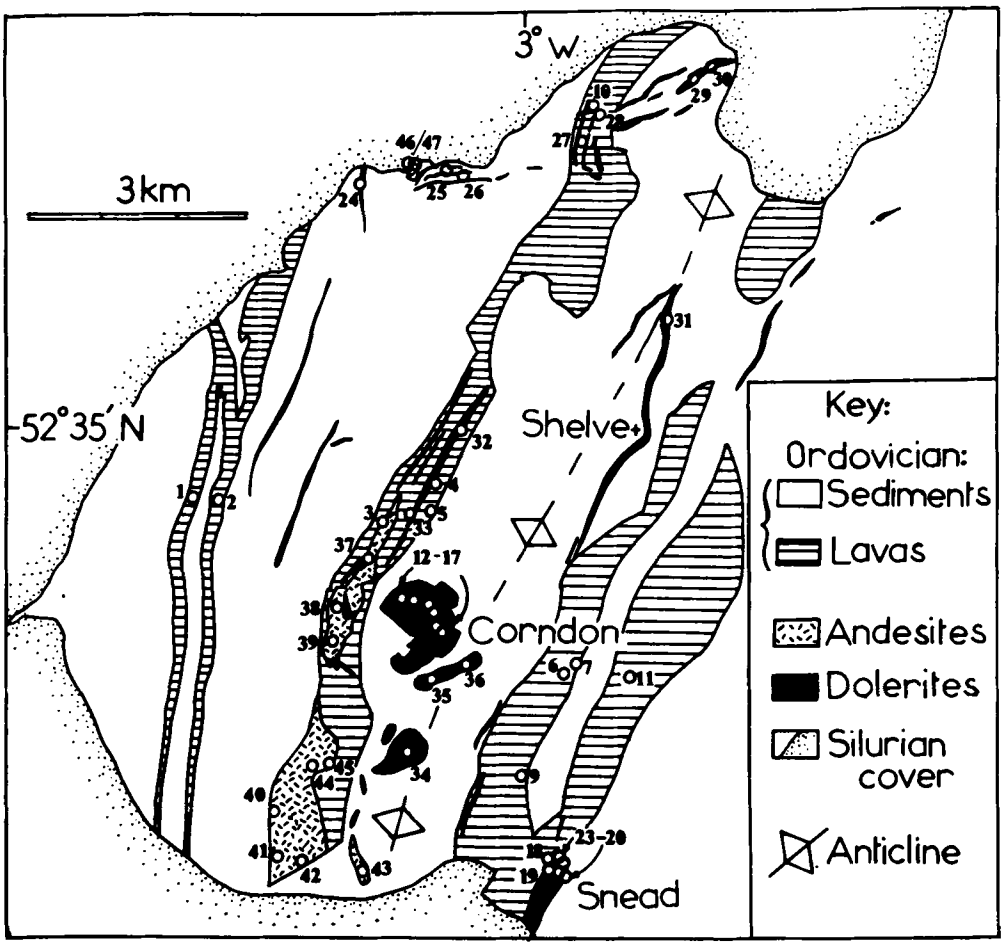


Figure 1. Geology and sampling sites in the Shelve (Ordovician) inlier, Welsh Borderlands. The geology is based on Pocock & Whitehead (1948) supplemented by data of Blyth (1944) for the igneous bodies. The regional location is shown in Fig. 2.

exposures with complementary thermal demagnetization studies would be necessary to evaluate the palaeomagnetic field at the time represented by these igneous events.

This paper reports the results of this work and also covers the palaeomagnetism of associated lavas and tuffs interleaved with Ordovician sediments assigned (from their graptolite fauna) to the Llanvirn and Caradoc Series. The survey has in addition been extended to the igneous rock units of the Berwyn Hills to complete palaeomagnetic coverage of Palaeozoic igneous rocks within central Wales and the Welsh Borderland (Fig. 2).

2 Geological outline and sampling

The Shelve inlier on the Shropshire–Montgomeryshire border comprises a complete sequence of Ordovician rocks from Arenigian to Caradocian* age (Lapworth & Watts 1894, 1910). Deposition ceased in Ashgill times when the region was deformed into broad NNE

* The Ordovician System is divided into the Tremadoc, Arenig, Llanvirn, Llandeilo, Caradoc and Ashgill Series, and the Silurian System is divided into the Llandovery, Wenlock and Ludlow Series.

trending open folds prior to deposition of Silurian (Upper Llandovery Series) sediments which now largely encircle the inlier. Deposition of the Ordovician sediments was interrupted by volcanism during the *Didymograptus bifidus* graptolite zone to produce the Stapeley Volcanic Group (palaeomagnetic sites 3–11, Fig. 1), and briefly by explosive volcanism producing tuffs during the *Dicranograptus clingani* zone of the Caradoc (sites 1 and 2, Fig. 1). The type Caradoc Series nearby was dated at 451 ± 21 Ma by Ross, Naeser & Izett (1976) using the fission-track method. At a later stage following post-Caradocian folding many andesite (sites 37–47, Fig. 1), dolerite (sites 12–36) and aplite bodies were intruded. The andesites are probably, for the most part, slightly older than the dolerites because they are intruded by the latter at one locality. The dolerites are quartz or hypersthene-bearing (Blyth 1944; Pocock & Whitehead 1948) and the most important representative of the latter group is intruded close to the apex of the Shelve anticline as the Corndon Hill phacolith (Blyth 1944; sites 12–17, Fig. 1). Near Snead, another intrusion, probably also a phacolith (sites 18–23, Fig. 1), was partially eroded prior to deposition of Upper Llandovery rocks (Blyth 1944); it is exposed to differing cooling levels by quarrying. A sill below the Corndon Phacolith is represented by sites 35 and 36, and the poorly exposed nearby Cwm-mawr picrite by site 34; this rock contains over 50 per cent olivine. Although most of the intrusions within the Shelve inlier are approximately conformable with bedding and occur as sills or phacoliths, dykes with east–west trend occur in the northern part of inlier (sites 24–30). These are a westward continuation of a dyke swarm within the Longmynd Precambrian inlier (Greig *et al.* 1968) and are believed to be part of the same consanguineous suite. The age of these intrusions within the Shelve inlier is restricted by geological relationships to the interval between late Ashgill and Upper Llandovery time (i.e. late Ordovician or early Silurian, *c.* 440 Ma).

In the Berwyn Hills a thick suite of Ordovician (Arenig to Ashgill Series) shales and mudstones is deformed into a major E–W anticline. Volcanic rocks are rare, the major exceptions being a tuff (the 'Cwnclwyd Ash', site B11, Fig. 2) extruded between Llandeilo and Caradoc times, and a higher tuff (the 'Pandy Ash', site B1). Site B11 cannot be taken as a precise stratigraphic marker because the Ordovician sequence is relatively condensed here; B1 occurs within the Caradocian succession. Deformation was followed by intrusion into Llandeilo sediments of an olivine dolerite sill at Hendre (sites B2–4). This dolerite is both chloritized and albitized and appears to postdate the regional cleavage (Wedd *et al.* 1922); no dip correction has accordingly been applied to these sites. The Hendre dolerite was associated with contemporaneous intrusion of keratophyric magma and may be related to a suite of keratophyre intrusions further to the southeast (sites B6–8 in quarry exposures of the Blodwell mass, B9 and 10 in the Bryn mass and B5 in the Llangynog mass, Fig. 2). These keratophyres are partly brecciated bodies representing volcanic necks cutting Caradocian sediments and, like the Hendre dolerite, experienced penecontemporaneous autometamorphism to form albite, epidote and chlorite. They are not seen to cut Ashgill or later sediments but all contacts with the Silurian (Lower Llandovery) are apparently faulted (Wedd *et al.* 1922), and since they appear to postdate the regional cleavage are considered to be post-late Silurian; they are unconformably overlain by (Lower) Carboniferous Limestone.

3 Palaeomagnetic results

3.1 METHOD

Five to ten 2.5-cm diameter cores, typically distributed over several metres of outcrop, were drilled at sites in the volcanics and intrusions (Figs 1 and 2) using a motorized drill, and

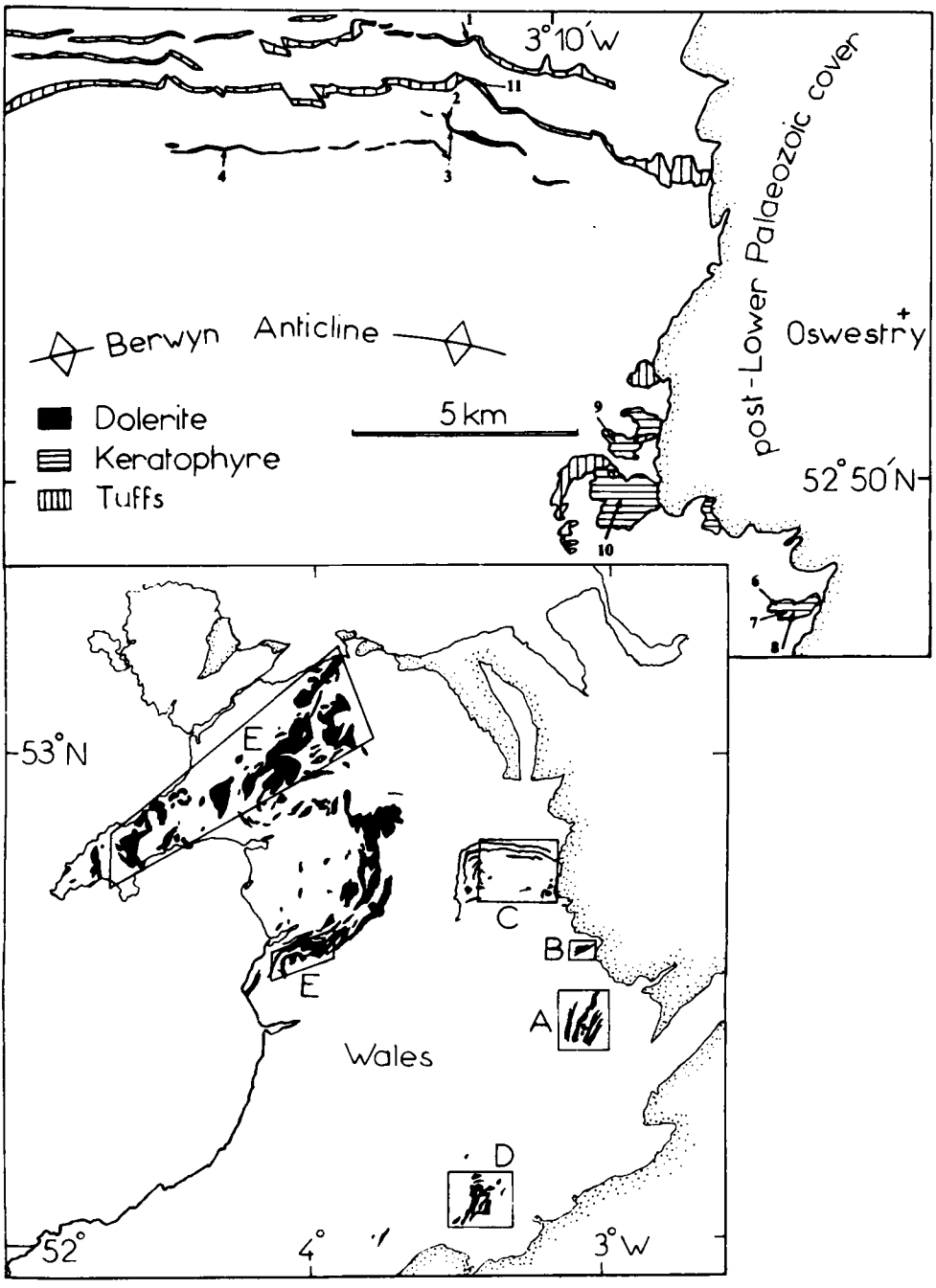


Figure 2. Outline geological map and sampling sites of the Berwyn Hills, Welsh Borderlands. The inset map shows the regions of this study in the context of the outcrop of Palaeozoic igneous rocks (black) in Central Wales and the Welsh Borderlands. Palaeomagnetic studies of these rocks are summarized: A, Shelve inlier, this study; B, Breidden Hill inlier (Piper & Stearn 1975); C, Berwyn Hills, this study; D, Builth Wells–Llandrindod Wells volcanic region, Piper & Briden (1973); E, North Wales Palaeozoic intrusions, Thomas & Briden (1976).

oriented by sights on the Sun and/or magnetic bearings in conjunction with topographic sights; because natural exposures in this region are usually heavily weathered much of the sampling was concentrated in outcrops exposed by quarrying. Magnetic moments were measured with a parastatic magnetometer system and to test stability of magnetizations all samples were progressively demagnetized in 500 Hz alternating field (af) up to several hundreds of oersteds (Oe) using a two-axis tumbler in steps of 50 or 100 Oe; thermal demagnetization techniques were further applied to test stability of the residual moments in the intrusive rocks and primary moments in some selected cores.

3.2 VOLCANIC ROCKS OF THE SHELVE INLIER

The Stapely Volcanics are very weakly magnetized and much material had to be rejected on these grounds; the samples yielding reliable results had intensities in the range 0.16–0.04 x 10^{-5} emu/gm. They typically exhibited small progressive fall in moments (Fig. 3) but in some cases became too weak to measure after treatment in fields of several hundreds of oersteds. In general, peak af fields of 150–250 Oe were found sufficient to produce

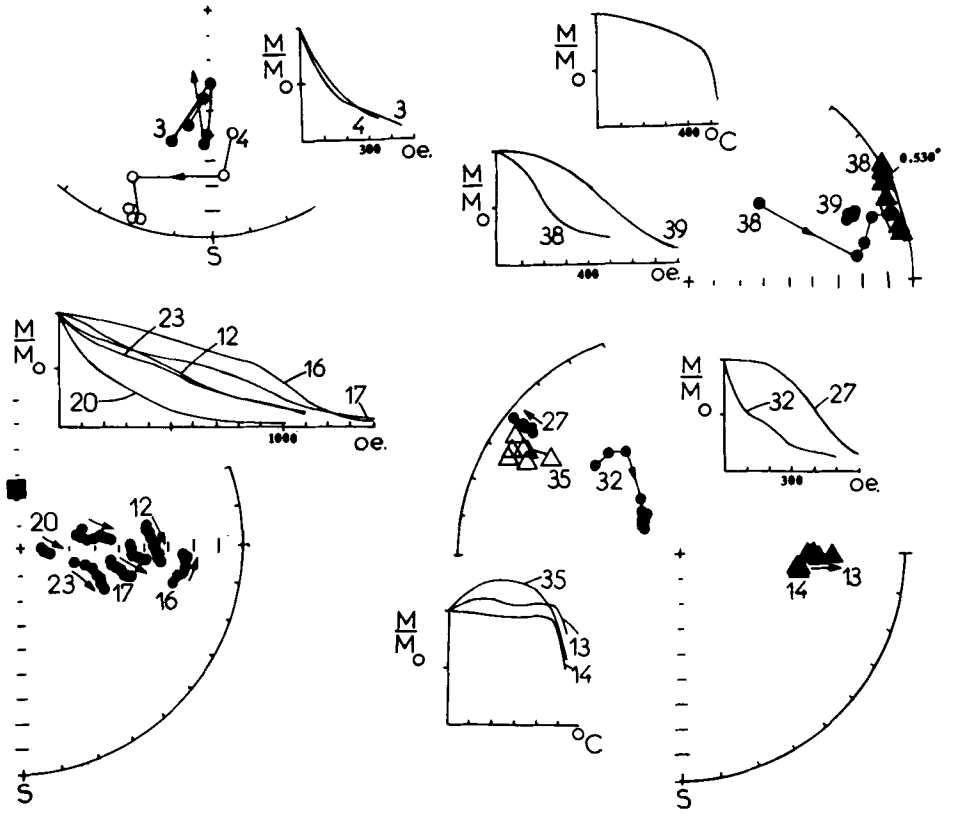


Figure 3. Behaviour of some representative palaeomagnetic samples from the Ordovician Shelve inlier with progressive af (circles) and thermal (triangles) demagnetization. Numbers refer to the site numbers of Tables 1–3 and open symbols are upper-hemisphere plots and closed symbols are lower-hemisphere plots. Behaviour of magnetic moments is expressed in terms of a fraction of the initial moment (M_0) prior to cleaning treatment. The arrows indicate directions of movement with progressive treatment and the projections are stereographic.

Table 1. Palaeomagnetic results, lavas and tuffs, Shelve inlier, (357° E, 52.5° N) tilt corrected and af cleaned.

Site no.	*N*	*D*	*I*	*R*	*k*	α95	NG ref.
1	10	201	29	8.48	6	21.7	32682984
2	7	182	29	5.98	6	27.3	32722984
3	7	186	20	6.09	7	25.4	33042981
4	8	185	43	7.54	15	14.6	33112984
5	6	165	19	5.23	6	28.8	33112982
6	6	160	36	5.60	13	19.6	33262957
7	6	185	51	5.83	28	12.9	33272958
8	5	167	25	4.55	9	27.2	33222933
9	7	171	20	6.39	10	20.3	33122939
10	7	320	−39	6.75	24	12.6	33333034
11	7	330	−18	6.09	7	25.5	33352956
Group means:							
(i) Llanvirnian Volcanics:							Palaeomagnetic pole
nine sites		167.3	30.9	8.60	20	11.8	10.0° E, 20.0° S ($d\Psi$ = 7.4, $d\chi$ = 13)
(ii) Caradocian Tuffs:							
six sites (see Section 3.1)		178.5	26.6	5.65	14	18.3	358.4° E, 23.4° S ($d\Psi$ = 10.8, $d\chi$ = 19)

N is the number of samples or sites, *D* the declination and *I* the inclination of the remanence vector with resultant *R* and precision parameter *k* (= $(N-1)/(N-R)$); α95 is the 95 per cent confidence circle about the mean direction and $d\Psi$ and $d\chi$ are the semi-axes of the oval of confidence about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively.

stable end-points of the magnetic vectors and higher fields produced either no further significant, or no further systematic, changes in direction. Thermomagnetic determinations yield Curie points between 460 and 580°C with comparable heating and cooling curves, confirming the evidence from demagnetization spectra (Fig. 3) that titanomagnetite is the principal remanance carrier. Site mean directions of magnetization significant at the 95 per cent confidence level are listed in Table 1 and the majority of the volcanics yield virtual geomagnetic poles lying near the equatorial South Atlantic and are thus reversely magnetized (Briden *et al.* 1973, Fig. 1), although two sites (10 and 11) from opposite sides of the Shelve anticline are magnetized approximately antiparallel to this main group.

The tectonic dip of the volcanics determined from interbedded sediments ranges from 40 to 56° in directions 277–304° E on the west limb of the Shelve anticline and 40 to 48° in directions 100–112° E on the east limb. Correction for dip assuming sites 10 and 11 to be reversely magnetized with respect to the remainder does not greatly affect the mean direction (D = 179°, I = 9° compared with D = 167°, I = 31°) but improves the overall precision from 4 to 20, a change significant at the 95 per cent confidence level and suggesting that the palaeomagnetic mean direction is prefolding (late Ordovician) in age. The mean south palaeomagnetic pole is 10° E, 20° S ($d\Psi$ = 7.4, $d\chi$ = 13.3) and presumably applies to the mid-Llanvirnian volcanic episode. Sites 1 and 2 from Caradocian tuffs cannot strictly be combined with the sites from the older Stapely Volcanic Group and are accordingly grouped with two tuffs of comparable age from the Berwyn Hills. (Sites B1 and 11, Section 3.3) and two Caradocian tuffs from the nearby Breidden Hill inlier (Piper & Stearn 1975). These six sites yield a poorly defined reversely magnetized palaeomagnetic

direction for Caradocian times, with a pole 358° E, 23° S ($d\Psi = 11°, d\chi = 20°$). Since these tuffs have different tectonic tilts we can again attempt a fold test and find that precision k improves from 4 to 14 after tilt correction significant at the 95 per cent confidence level (McElhinny 1964); this test coupled with the observation that the derived palaeomagnetic directions are entirely removed from Tertiary and Recent geomagnetic field directions confirms their ancient origin.

3.2 ANDESITE AND DOLERITE INTRUSIONS OF THE SHELVE INLIER

With progressive af treatment most andesite samples move to easterly or westerly declinations with shallow inclinations and with additional thermal demagnetization these directions are retained until the demagnetization curves show a sharp fall in moment above 500°C (Fig. 3). Site mean statistics are given in Table 2 where it is clear that a precise reversal of magnetization is recorded within these intrusions. In particular within the large and complex andesite intrusion west of Corndon Hill (Fig. 1) sites 37–39 and 41 have opposite polarity to sites 40, 42 and 43, and it seems probable that the two sites in close proximity, 44 and 45, which yield group mean directions unrelated to the other sites, have recorded intermediate transitional directions of the palaeofield. The andesite body at Lower Wood (NG Ref. 33093025) was heavily sampled in quarry outcrops covering most of the present level of exposure, but it seems unlikely that many of the samples have faithfully recorded the palaeofield direction of the time of igneous cooling because they retain steep positive magnetizations with progressive af treatment (site 47) although four samples, including the highest coercivity members of the group, move to shallow westerly directions with progressive treatment (site 46, Table 2) and stabilize with directions comparable to sites 40, 42 and 43. The group mean of eight sites assuming dual polarity, is $D = 100°$, $I = 2°$ with a palaeomagnetic pole at 78° E, 5° S ($d\Psi = 6.7°, d\chi = 13.4°$).

Dolerite samples from the major phacolith bodies of Corndon and Snead exhibit slight systematic movement of magnetic vectors with progressive treatment (Fig. 3), with some movement over the whole range of treatment. Since this movement is in most cases close to

Table 2. Palaeomagnetic results, Andesite intrusions, Shelve inlier, (357° E, 52.5° N) af and thermally cleaned.

Site	N	D	I	R	k	$\alpha 95$	M_0 (X 10^{-5} emu/gm)	NG ref.
37	6	111	5	5.94	84	7.4	1.77–0.58	33032975
38	7	79	−9	6.84	37	10.0	0.06–0.02	32982968
39	7	75	16	6.85	39	9.8	1.11–0.63	32952966
40	5	290	5	4.94	70	9.2	0.04–0.02	32862942
41	8	116	15	6.01	4	34.6	0.19–0.03	32872938
42	6	286	−9	5.55	11	20.9	0.31–0.07	32882934
43	5	275	−3	4.39	7	32.4	0.07–0.01	33002934
44*	6	304	58	5.65	14	18.4	0.16–0.08	32932952
45*	7	174	−27	6.71	20	13.7	0.03–0.01	32912948
46	4	290	16	3.73	11	28.8	3.39–1.59	33093025
47*	15	336	66	14.28	19	8.9	12.80–0.05	
Group mean:							Palaeomagnetic pole	
	eight sites	100.3	2.3	7.61	18	13.4	78.2° E, 5.3° S ($d\Psi = 6.7, d\chi = 13.4$)	

* Sites excluded from group mean calculation.

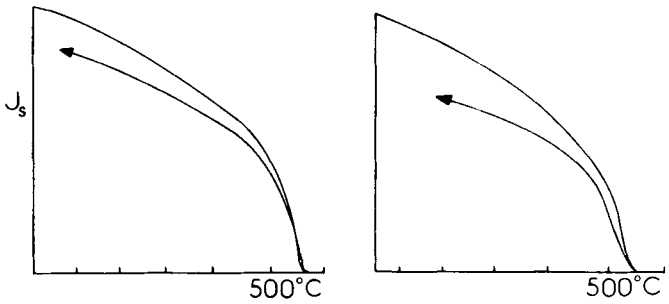


Figure 4. Typical thermomagnetic determinations (saturation magnetization Js versus temperature) from dolerite samples, Shelve inlier.

a great-circle line connecting the sample direction with present geomagnetic field direction it may represent subtraction of a present-day field component, although as some movement occurs, over the whole range of field treatment available (0–1400 Oe), it must include high coercivity components, and it is not possible to exclude aquisition of different blocking temperature components in a rapidly changing ancient field. However, this behaviour does not appreciably affect the selection of a group mean direction and the palaeomagnetic pole for the Corndon Hill phacolith, for example, lies at 67° E, 15° N when calculated from 200–400 Oe demagnetization data, and at 69° E, 15° N when calculated from 1400 Oe data. Progressive thermal demagnetization (Fig. 3) demonstrates that the af method has been successful in removing components of secondary magnetization, and that magnetic moments reside in single high blocking temperature components in titanomagnetite; the Curie points suggested by thermal demagnetization spectra are confirmed by Js–T determinations yielding Curie points of 550–570°C or near to that of unsubstituted magnetite (Fig. 4). Opaque petrologic studies suggest that this highly stable remanence resides in abundant titanomagnetite variably altered to a range of oxidation products that characterize both the advanced stages of deuteric oxidation and (lower temperature) hydrothermal alteration (Ade-Hall, Palmer & Hubbard 1971). Whilst some homogeneous titanomagnetites are apparently unaffected, other grains in juxtaposition and exhibiting subsolidus exsolution have a high density of spinel rods and hematite and rutile within the ilmenite lamellae; the latter alteration is also present as ramifying veins isolating unaltered grain interiors. This advanced stage of alteration very rarely takes place at deuteric temperatures in basic intrusions (Ade-Hall & Lawley 1970) from which it also appears to be distinguished by its sporadic and irregular development; it may result from the concentration of residual magmatic fluids near the impermeable roof of the body which forms the presently exposed level (Blyth 1944). Piper & Stearn (1975) found similar hydrothermal effects concentrated at the upper and lower margins of the nearby Breidden Hill dolerite body, although in this case the magnetic response (leading to lower magnetic intensities and stabilities) was different, possibly resulting from the critical reactions taking place at lower temperatures.

Although individual sites within the Snead phacolith show compact within site grouping (Table 3), the separate sites exhibit considerable variation in magnetic inclination with small range in declination. Since they define a field which also includes site mean directions of the Corndon body, it is pertinent to examine the significance of this variation. The exposed area of the Corndon body lies near the inferred roof of the phacolith, while Snead had been exposed to differing levels by quarrying and the magnetic inclination appears to increase towards the base of the exposed section through the phacolith in the approximate sequence of sites 20, 22 to 18, 19 and 23. Since a body with the thickness of Snead (~ 200 m) would

Table 3. Palaeomagnetic results, Dolerite intrusions, Shelve inlier, (357° E, 52.5° N) af and thermally cleaned.

Site no.	*N*	*D*	*I*	*R*	*k*	α95	M_0 (× 10^{-5} emu/gm)	NG ref.
Corndon Hill Phacolith (357.0° E, 52.6° N)								
12	7	96	41	6.98	352	3.2	5.76–3.48	33032972
13	7	94	38	6.98	246	3.9	7.26–4.33	33052972
14	7	94	37	4.95	82	8.5	147.50–1.91	33082971
15	6	110	27	5.98	240	4.3	68.20–6.14	33102969
16	7	81	45	6.55	13	17.2	55.20–7.12	33102966
17	7	89	53	6.95	116	5.6	6.13–2.82	33102965
Mean, Corndon Phacolith (six sites)							Palaeomagnetic pole	
		94.8	40.3	5.90	50	9.5	69.0° E, 15.3° N ($d\Psi$ = 7.0, $d\chi$ = 11.6)	
Snead Phacolith (357.0° E, 52.5° N)								
18	5	87	41	4.78	18	18.5		
19	9	125	18	8.93	108	4.9	0.28–0.15	33242933
20	7	120	71	6.77	26	12.1	0.76–0.80	33252933
21	5	115	56	4.88	33	13.4	32.30–27.81	33272934
22	7	102	76	4.37	295	3.5	31.30–2.89	33272935
23	6	113	26	5.30	7	26.9	27.30–7.42	33272936
Mean, Snead Phacolith (six sites)							Palaeomagnetic pole	
		111.1	48.8	5.51	10	22.0	52.9° E, 11.8° N ($d\Psi$ = 19.1, $d\chi$ = 29.0)	
Miscellaneous dolerite intrusions								
24	8	299	7	6.12	4	33.3	0.37–0.04	33023022
25	7	294	−4	6.88	52	8.4	69.00–1.75	33123027
26	8	299	−14	7.12	8	20.8	3.39–0.14	33143026
27	7	310	0	6.77	26	12.0	15.40–0.03	33353035
28	5	305	−11	4.46	7	30.0	11.80–0.06	33313004
29	5	306	−33	4.87	31	13.9	0.04–0.02	33463036
30	8	315	−27	7.69	22	11.9	0.49–0.02	33473038
31	7	106	77	6.67	18	14.4	0.03–0.02	33433005
32	7	124	51	6.86	44	9.2	0.08–0.03	33122990
33	6	232	79	5.05	5	32.2	3.04–0.07	33022980
34*	7	96	65	6.83	36	10.2	151.80–1.91	33072949
35	7	314	−26	6.92	79	6.8	0.34–0.13	33132958
36	7	323	60	6.85	41	9.5	0.87–0.20	33072957

* Picrite intrusion.

Group mean directions: (see Section 3.2):							Palaeomagnetic poles
(i) Snead Phacolith plus miscellaneous steep directions:							
	10	113.2	58.5	9.14	10	15.7	45.8° E, 18.4° N ($d\Psi$ = 17.2, $d\chi$ = 23.3)
(ii) Other miscellaneous dolerites:							
	8	305.0	−13.6	7.73	26	11.1	234.2° E, 14.5° S ($d\Psi$ = 5.8, $d\chi$ = 11.3)
(iii) Shelve–Longmynd E–W dykes:							
	20	309.8	−14.7	19.29	27	6.4	229.6° E, 16.4° S ($d\Psi$ = 3.4, $d\chi$ = 6.6)

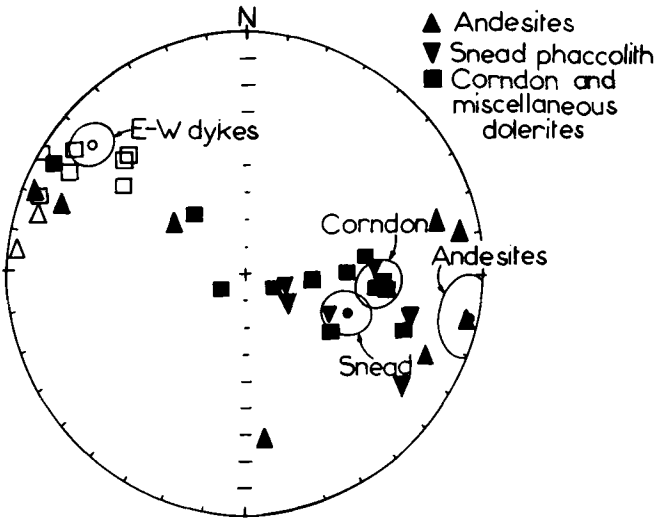


Figure 5. Cleaned *in situ* directions of magnetization and 95 per cent confidence circles for intrusive bodies of the Shelve inlier.

only have taken a period of the order of 10^2yr to cool through the Curie point (Jaegar 1964) if intruded at shallow crustal levels, the palaeomagnetism must have recorded a rather rapid polar movement, possibly part of a geomagnetic field reversal. Evidence for field reversal is corroborated by data from the andesites and other minor dolerite intrusions (Fig. 5). Indeed the remaining minor dolerite intrusions (Table 3) fall into two groups: certain sills near the centre of the inlier and the Cwm-mawr picrite have steep positive directions of magnetization analogous to some directions of the Snead phacolith, while the remainder, comprising with one exception, the east–west trending dykes of the northern part of the inlier, have shallow northwesterly directions (Fig. 5). These site means have been grouped accordingly in Table 3.

The east–west dolerite dykes (sites 25, 26, 28–30) are the western extension of a swarm within the Longmynd Hills (Greig *et al.* 1968, p. 57). Lomax & Briden (1977) interpreted the palaeomagnetic directions from this swarm as Cambrian or Late Precambrian because they were unlike any post-Cambrian directions identified from Britain. However, the directions are identical to the dykes of this area from rocks of undisputed late Ordovician/early Silurian age and confirm the correlation of Greig *et al.* (1968). The results of the present study are combined in Table 3 with the results from eight dykes (aberrant site 133 excluded) and four dykes contacts of Lomax & Briden (1977) to derive a group mean for this dyke swarm.

A complete analysis of directional data from all of these intrusions involves assessment of two points. First, whether any of the intrusions might have been emplaced prior to regional folding, and second, whether the moderate to steep magnetic inclinations all represent the record of a polarity transition, as may be the case for the Snead data. The first points must be considered as a possible explanation of the anomalous directions because Blyth (1944) noted slight relative movement of joint blocks within the Corndon body that might represent post-emplacement folding. However, any significant post-cooling tectonism can be discounted by three arguments: (i) the andesites probably represent the oldest intrusive episode (Section 2) and the approximate agreement of data for these rocks (with dual polarity) with the east–west dolerite dykes in the northern part of the inlier which in turn

agree precisely with analogous dykes in the Longmynd region is effectively destroyed by applying a tilt correction; however, since the andesites all occur along one limb of an anticline the dual polarity alone cannot resolve this point, and there remains some uncertainty whether these rocks were intruded prior to folding (if corrected for dip the direction of magnetization and pole position are moved to $D = 100°$, $I = 42°$, and 64° E, 14° N, respectively); (ii) the Corndon Hill phacolith lies close to the axis of the Shelve anticline and a correction for the dip of intruded sediments would be very small, indeed correcting the slight aberrant site 15 from the east side of the phacolith would remove the direction of magnetization further from the remaining sites; (iii) variation of magnetic direction within the Snead body occurs with depth in the intrusion but at similar structural contours with the overlying conformable sediments, and correction of directions for these sites for the regional dip, immediately to the west (60° in a direction 280° E) would in turn remove them from the Corndon directions.

Two lines of evidence suggest that the slight differences in mean directions between the intrusions (Table 3) cannot be explained in terms of transitional geomagnetic field directions. First, this would require that they cooled within a short part of a reversal cycle (*c.* 5×10^3yr) although they formed as hypabyssal bodies emplaced at high crustal levels over an area of 35 km^2. Secondly, the magnetic intensities of the rocks with steeper magnetic directions are an order higher (Table 3) than in the east–west dolerite dykes and andesites which include directions of both polarities: the reverse would be typical if they were transitional field directions. It is thus probable that all the palaeomagnetic poles from these intrusions are true records of ancient poles although they represent a new problem of interpretation because they are quite unlike other poles reported from British Palaeozoic rocks (Section 4; Briden *et al.* 1973).

3.3 IGNEOUS ROCKS OF THE BERWYN HILLS

Tuffs B1 and 11 exhibit small change in direction of NRM with progressive af treatment and stabilize with 50–100 per cent of total remanence remaining. The weakly magnetized Hendre dolerite and keratophyre sites from Blodwell (B6–8) exhibit comparable behaviours and stabilize with shallow southerly directions after af treatment in low fields (Fig. 6,

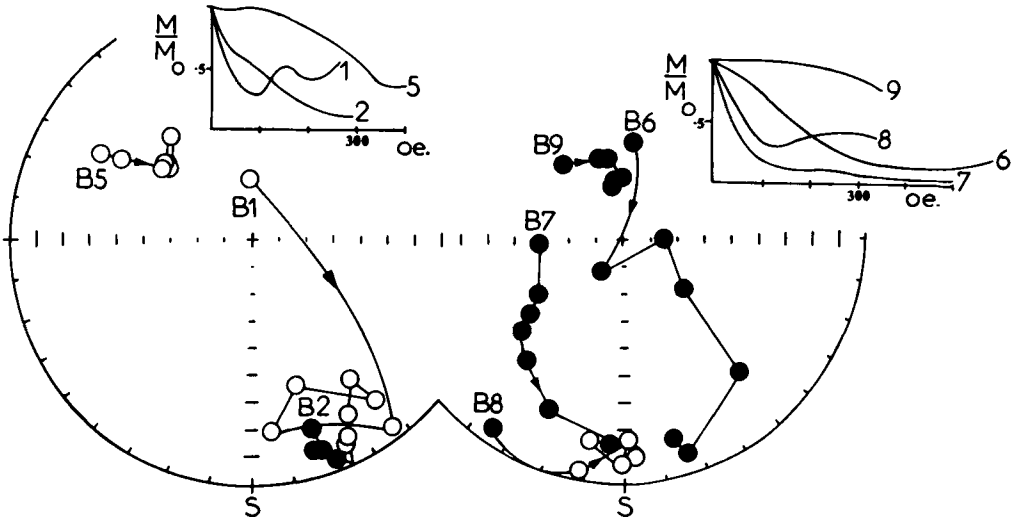


Figure 6. Behaviour of some palaeomagnetic samples from the Berwyn Hills with progressive af treatment. Symbols are as in Fig. 3.

Table 4. Palaeomagnetic results, Berwyn Hills (356.8° E, 52.8° N).

Site	*N*	*D*	*I*	*R*	*k*	$\alpha 95$	Grid ref.	M_0 (× 10^{-5} emu/gm)
B1 (Pandy Tuff)*	8	181	11	7.74	27	11.0	31983364	0.38–0.02
					(VGP, 335.8° E, 31.7° S)			
B2	6	175	−1	5.73	19	16.0	31913347	0.06–0.02
B3	7	184	10	6.70	20	13.8	31913346	0.12–0.03
B4	6	212	15	5.32	7	26.5	31403342	0.03–0.02
B5	7	328	−64	6.68	19	14.3	30473273	0.10–0.03
B6	7	192	24	6.51	12	18.0	32573231	0.13–0.04
B7	5	193	14	4.47	8	29.7	32583230	11.78–0.15
B8	8	180	−10	7.64	20	12.8	32603229	1.08–0.26
B9	7	5	77	6.95	111	5.8	32223730	0.16–0.04
B10	7	352	69	6.84	37	10.0	32243258	0.36–0.05
B11 (Cwnclwyd Tuff)*	4	153	6	3.94	47	13.6	31973353	1.80–0.03
					(VGP, 28.6° E, 29.5° S)			

* Tilt corrected; VGP = virtual geomagnetic pole.

Group means:							Pole positions	
							° E	° S
(i) Hendre dolerite (B2–4)								
	19 samples	189.1	8.5	17.0	9	11.8	346.0	32.3
(ii) Blodwell keratophyre (B6–8)								
	20 samples	186.9	7.5	17.9	9	11.6	348.6	33.0

Table 4), with the exception of B7 from the deepest exposed level of this intrusion which exhibits two high coercivity components of magnetization. One of these lies near the present magnetic field and may be either a VRM or a PTRM because it lies close to directions from B9 and 10 in nearby keratophyre bodies; in this case it is certainly secondary because it could be removed by high af treatment in fields of 600–1200 Oe or thermal treatment above 400°C. The remaining keratophyres (B5, 9 and 10) yield steeper directions stable to high af fields. Site B5 is stable to progressive thermal demagnetization, seven samples grouping at $D = 334°$, $I = -44°$ ($k = 9$) after cleaning at 500°C, but B9 and 10 lie near the present geomagnetic field and only B9 retained a measurable moment with progressive thermal treatment giving a mean $D = 310°$, $I = 71°$ at 450°C.

To derive group mean directions of magnetization (Table 4) stable directions from the Hendre dolerite and Blodwell keratophyre were combined at the sample level because sites are either in close proximity to one another, or derive from the same igneous event.

4 Interpretation of the Palaeomagnetic results

4.1 THE TIME SEQUENCE OF PALAEOMAGNETIC POLES

The Stapely Volcanic Group, belonging to the *Didymograptus bifidus* graptolite zone are comparable to the Eycott Group of the northern Lake District (Briden & Morris 1973, *Didymograptus bifidus* zone or slightly younger) and are slightly older than the Builth Volcanic Series (Piper & Briden 1973, *Didymograptus murchisoni* zone). With the exception of two sites, the Stapeley Group is reversely magnetized like the Builth Volcanic Series and of opposite polarity to the Eycott Group; a polarity sequence R → N → R is thus implied by the collective results. The pole positions for these studies are all very similar and coupled with the differing tectonic corrections of the separate collections, define the Lower

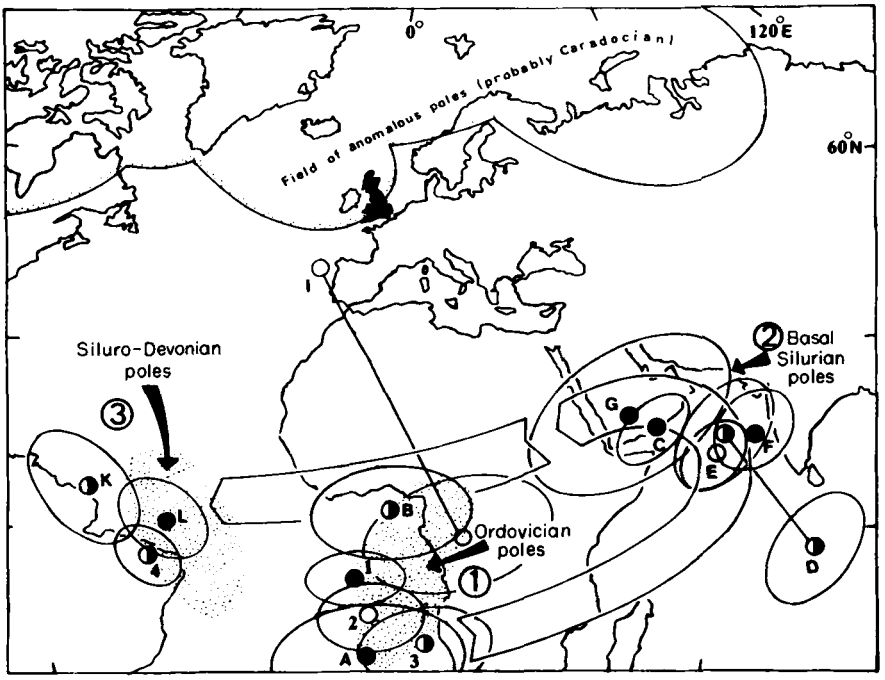


Figure 7. The rapid apparent polar movement of Late Ordovician–Early Silurian times defined by the data of Tables 1–5. The palaeomagnetic poles are plotted with 95 per cent confidence ovals, closed symbols are reversed poles, open symbols are normal. The pole from the andesites of the Shelve inlier is plotted before and after dip correction because of uncertainty whether or not this should be applied. Lettered poles are listed in Table 5 and other important poles (Briden *et al.* 1973) are numbered: (1) Eycott Group, northern Lake District; (2) Builth Volcanic Series, Central Wales; (3) Stapeley Volcanic Group (this study); (4) Lower Old Red Sandstone (Upper Silurian/Lower Devonian) lavas and sediments, Midland Valley of Scotland. The field of anomalous palaeomagnetic poles, of known or probable Ordovician age is also indicated (Briden & Thomas 1976; Piper *et al.* 1978).

Ordovician palaeomagnetic pole (Fig. 7). The Caradocian tuffs from the Welsh Borderlands define a similar, but slightly shallower, geomagnetic field axis than the Lower Ordovician rocks.

The palaeomagnetic poles from the andesite and dolerite intrusions are removed by nearly 90° from poles regarded as typical of Ordovician and Silurian times (Section 1). Laboratory and field evidence for stability is discussed in Section 3, and a regional tectonic explanation can also be discounted because not only is the post-Silurian tectonic history of the region a simple one with large rotations about horizontal or vertical axes being inconceivable, but also such movements would have influenced the Lower–Middle Ordovician data which demonstrably agree with other British data. In the context of their geological age, the andesites would define the oldest pole and the dolerites the youngest poles although it is difficult to arrange this latter group in a time sequence on geological evidence. The Snead intrusion is pre-Upper Llandovery in age and the time sequence of poles is clarified by results from the nearby Breidden Hill dolerite (356° 57′ E, 52° 43′ N), a hypersthene dolerite similar to the Corndon phacolith. The observation that the Shelve dolerites postdate the end-Ordovician tectonism raises the question as to whether this body might also have been intruded after folding of the local Ordovician rocks; it cannot be precisely contemporaneous with the Shelve dolerites because it is magnetized in the opposite (normal) sense (Piper & Stearn 1975). There is also geological evidence that it is younger because one member of the

dolerite suite intrudes Middle–Upper Llandovery sediments (Watts in discussion to Blyth 1944; Whittard 1932); unfortunately the critical relationships (Whittard 1932) are no longer exposed for palaeomagnetic study. Piper & Stearn were therefore incorrect in applying a tilt correction to the palaeomagnetic results for post-Caradoc folding because the inlier is pre-Llandovery and is partially overlain by unconformable Llandovery rocks, the pole derived from *in situ* directions lies at 166° E, 43° S ($D = 40, I = -84, k = 17$) although a component of end-Silurian folding may still be unaccounted for. Piper & Stearn found aberrant directions of magnetization in the post-Caradoc, pre-Upper Llandovery Moel-y-Golfa andesite from the same area: detailed analysis with thermal demagnetization has yielded an underlying stable direction $D = 281$, $I = -47°$ and pole at 242° E, 16° S believed to date from time of eruption; this direction is corrected for dip measured at outcrop, and the pole lies close to results from intrusions of the Shelve area (Fig. 7, Table 5).

4.2 LOWER PALAEOZOIC APPARENT POLAR MOVEMENTS

The collective data define a movement from the typical Ordovician pole position for Britain in late Ordovician and Lower Silurian time (Fig. 7). The timing of this event is further defined as post-late Ashgill by other British data (Table 5) and may therefore have been restricted entirely to Silurian times. The next youngest pole is derived from Silurian lavas of Somerset and Gloucestershire (Piper 1975) where both polarities are represented and a fold test confirms that remanence relates to primary deuteric petrogenesis. It was originally believed that these lavas spanned the junction between Upper Llandovery and Lower Wenlock times, but new palaeontologic data from the brachiopod fauna (Bassett 1974; M. L. K. Curtis, private communication) show that they belong entirely to the Wenlock Series. Referred to the absolute timescale this pole would thus be slightly younger than 424 Myr (Lambert 1971, $\lambda\, Rb^{87} = 4.85 \times 10^{10}$yr), and is close to a group of poles assigned to the Upper Silurian–Lower Devonian interval (Briden *et al.* 1973). To this latter group should be added a further pole (141° E, 2° S) from the Lorne Plateau lavas dated 421 ± 13 Myr ($\lambda\, Rb^{87} = 4.85 \times 10^{10}$yr) (Latham & Briden 1975). The collective information currently suggests slight polar movement from *c.* 425 to 390 Myr (Fig. 7). The palaeomagnetism of Middle–Upper Devonian times is less well understood, but we note in passing that the poles for the Hendre dolerite and Blodwell keratophyre (Table 4) are in close agreement with results from Upper Old Red Sandstone (Upper Devonian–Lower Carboniferous) sediments of the Bristol District (palaeomagnetic pole 158° E, 32° N, Morris *et al.* 1973).

4.3 EXPLANATION OF THE LATE ORDOVICIAN/LOWER SILURIAN POLAR MOVEMENT

Of special geomagnetic or tectonic interest is the large movement of the dipole axis defined by the Llandovery age results of this study, and confined by reference to absolute age dates (Lambert 1971) to a period of less than 10 Ma between *c.* 445 and 425 Ma (Table 5). Since the earlier (and better defined) portion of this movement represents a change predominantly in magnetic *declination* it could be explained in terms of a rotation of the British region largely *in situ*, and it is noteworthy that the movement coincides with the major episode of tectonic activity at the end of Ordovician times within the paratectonic Caledonides of Britain.

Because Britain lay in low latitudes in Lower Paleozoic times its position was sensitive to rotational movements. It can be seen from Fig. 8 that, when the mean pole positions of Fig. 7 are converted into their equivalent palaeolatitudes, they can be explained in terms of a

Table 5. Palaeomagnetic poles defining the Late Ordovician – Early Silurian polar movement relative to Britain.

Pole	Rock unit	Geologic age System/Series		Stratigraphic age	Approximate absolute age (Ma)	Pole position		Polarity	Ref.
A	Tuffs, Welsh Borderlands			Caradoc	447–451	358° E	23° S	R	This study
B	Dolerite intrusions, Builth District	Ordovician	Caradoc	Late Caradoc–early Ashgill		2	2° N	M	Piper & Briden (1973)
C	Cautley Fell Rhyolite		Ashgill	Middle–Upper Ashgill		335	48° S	R	Author, unpublished
D	Andesite intrusions, Shelve			(post-Caradoc)		78	5	M	This paper
E	Moel-y-Golfa Andesite			(post-Caradoc)		62	16° N	N	See text
F	Corndon Hill Phacolith			< pole D		69	15	R	This paper
G	Snead phacolith and miscellaneous dolerite intrusions		Llandovery	(pre-Upper Llandovery)		46	18	R	This paper
H	E–W dykes, Shelve–Longmynd area	Silurian		(pre-Upper Llandovery)		50	16	N	Table 3
J	Breidden Hill dolerite			(post-Upper Llandovery)		346	43	R	See text
K	Lavas, Somerset and Gloucestershire		Wenlock	Wenlock	>427	305	7	M	Piper (1975)
L	Lorne Plateau Lavas, Scotland		Ludlow (?)		421	321	2	N	Latham & Briden (1975)

The age estimates are based on Lambert (1971) and on data quoted in references, Rb–Sr dates use Rb^{87} half-life of 4.85×10^{10}yr. It is not definitely established that pole J belongs to this sequence. The unpublished result C is based on 18 af cleaned samples from two sites; the relevant data for the other results is given in references quoted.

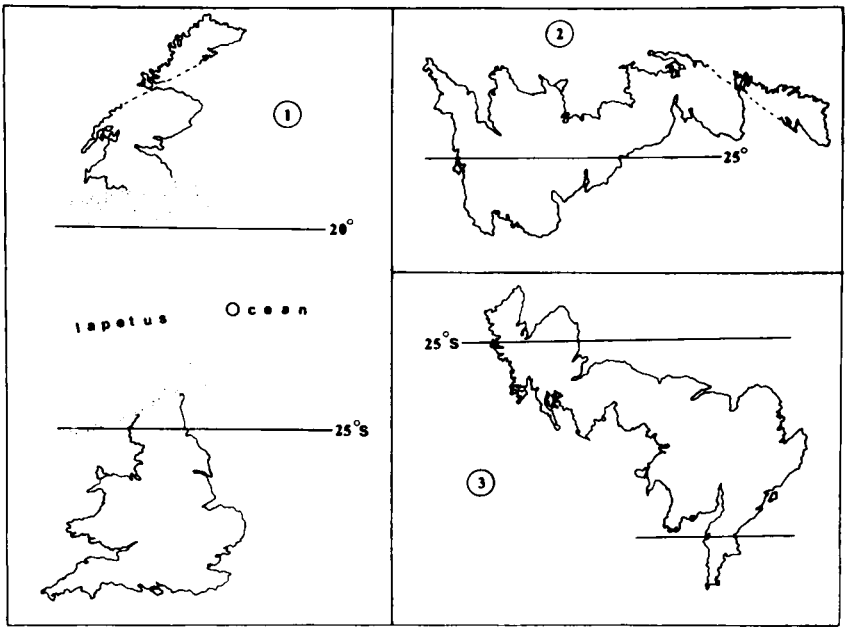


Figure 8. The apparent polar movement of Fig. 7 translated into the palaeolatitudes of Britain in Ordovician and Silurian times, and defined by the means of the three groups of poles illustrated in Fig. 7. The width of the Iapetus ('Proto-Atlantic') Ocean is estimated from Llanvirn–Llandeilo (Lower–Middle Ordovician) palaeomagnetic data (Piper 1978) and there is as yet no evidence that this ocean was still in existence in Late Ordovician and Silurian times.

clockwise rotation of Britain in latest Ordovician or earliest Silurian times followed by an anticlockwise rotation in middle Silurian times. Only slight latitudinal movement is implied by the poles and this is especially true if the andesites were magnetized prior to folding (Fig. 7); the Breidden Hill dolerite pole may then record the rapid Silurian clockwise rotation. If all individual poles are taken into consideration more latitudinal movement cannot be excluded, although part of this latter problem is related to uncertainties in tectonic tilt corrections (Briden *et al.* 1973) since almost all the source rocks have been subjected to late Ordovician and/or late Silurian tectonism.

It is tempting to correlate these rotations with final closure of the Iapetus or 'Proto-Atlantic' ocean during formation of the Caledonides. The palaeomagnetic evidence cannot yet resolve this point but we note that Lower–Middle Ordovician declinations on either side of the Iapetus suture are comparable (Briden *et al.* 1973; Piper 1977, Fig. 8) and imply that the two margins were parallel prior to the movements described here. These movements have not yet been identified elsewhere, no doubt because contemporaneous data spanning this relatively short interval do not yet exist. However, they do serve to emphasize that apw paths become increasingly more complex as more data become available, and it is no longer a necessary or sufficient justification for older proposed apw paths to argue that the rates of movement they involve are comparable to Mesozoic and Tertiary rates. This point is emphasized by the discovery of anomalous geomagnetic fields in Ordovician times (Thomas & Briden 1976; Piper *et al.* 1977).

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References

Ade-Hall, J. M. & Lawley, E. A., 1970. An unsolved problem – opaque petrological differences between Tertiary basaltic dykes and lavas, in *Mechanisms of igneous intrusion*, pp. 217–230, eds Newall, G. & Rast, N., Gallery Press, Liverpool.

Ade-Hall, J. M., Palmer, H. C. & Hubbard, T. P., 1971. The magnetic and opaque petrological response of basalts to regional hydrothermal alteration, *Geophys. J. R. astr. Soc.*, **24**, 137–174.

Bassett, M. G., 1974. Review of the stratigraphy of the Wenlock Series in the Welsh Borderland and South Wales, *Palaeontology*, **17**, 745–777.

Blyth, F. G. H., 1944. Intrusive rocks of the Shelve area, South Shropshire, *Quart. J. geol. Soc. Lond*, **99**, 169–204.

Briden, J. C. & Morris, W. A., 1973. Palaeomagnetic studies in the British Caledonides – III. Igneous rocks of the northern Lake District, England, *Geophys. J. R. astr. Soc.*, **34**, 27–46.

Briden, J. C., Morris, W. A. & Piper, J. D. A., 1973. Palaeomagnetic studies in the British Caledonides – VI. Regional and global implications, *Geophys. J. R. astr. Soc.*, **34**, 107–134.

Greig, D. C., Wright, J. E., Haines, B. A. & Mitchell, G. H., 1968. Geology of the country around Church Stretton, Craven Arms, Wenlock Edge and Brown Clee (explanation of one-inch Geological Sheet 166 new series), *Mem. geol. Surv. U.K.*, 379 pages.

Jaegar, J. C., 1964. Thermal effects of intrusions, *Rev. Geophys. Space Sci.*, **2**, 443–466.

Lambert, R. St. J., 1971. The pre-Pleistocene Phanerozoic time scale – a review, in *The Phanerozoic time scale – a supplement*, ed. Harland, W. B., *Spec. Publ. geol. Soc. Lond.*, 594–596.

Lapworth, W. & Watts, W. W., 1894. Geology of South Shropshire, *Proc. geol. assoc.*, **13**, 297–328.

Lapworth, W. & Watts, W. W., 1910. Shropshire, in *Geology in the field*, Jubilee volume Geological Association, part 4, p. 749.

Latham, A. G. & Briden, J. C., 1975. Palaeomagnetic field directions in Siluro–Devonian lavas of the Lorne Plateau, Scotland, and their regional significance, *Geophys. J. R. astr. Soc.*, **43**, 243–252.

Lomax, K. & Briden, J. C., 1977. Palaeomagnetic studies of the Longmyndian and other British late Precambrian/early Palaeozoic rocks and their regional tectonic implications, *J. geol. Soc. Lond.*, **133**, 5–21.

McElhinny, M. W., 1964. Statistical significance of the fold test in palaeomagnetism, *Geophys. J. R. astr. Soc.*, **8**, 338–340.

Morris, W. A., Briden, J. C., Piper, J. D. A. & Sallomy, J. T., 1973. Palaeomagnetic studies in the British Caledonides – V. Miscellaneous new data, *Geophys. J. R. astr. Soc.*, **34**, 69–106.

Piper, J. D. A., 1975. Palaeomagnetism of Silurian Lavas of Somerset and Gloucestershire, England, *Earth planet. Sci. Lett.*, **25**, 355–360.

Piper, J. D. A., 1978. Palaeomagnetism and palaeogeography of the Southern Uplands Block in Ordovician times, *Scot. J. geol.*, in press.

Piper, J. D. A. & Briden, J. C., 1973. Palaeomagnetic studies in the British Caledonides – I. Igneous rocks of the Builth Wells–Llandrindod Wells Ordovician inlier, Radnorshire, Wales, *Geophys. J. R. astr. Soc.*, **34**, 1–12.

Piper, J. D. A. & Stearn, J. E. F., 1975. Palaeomagnetism of the Breidden Hill (Palaeozoic) inlier, Welsh Borderlands, *Geophys. J. R. astr. Soc.*, **43**, 1013–1016.

Piper, J. D. A., McCook, A. S., Watkins, K. P., Brown, G. C. & Morris, W. A., 1978. Palaeomagnetism and chronology of Caledonian igneous episodes in the Cross Fell inlier and North Lake District, *J. Geol.*, in press.

Pocock, R. W. & Whitehead, T. H., 1948. *British regional geology: the Welsh Borderland*, 31 pages, HMSO, London.

Ross, R. J., Naeser, C. W. & Izett, G. A., 1976. Apatite fission-track dating of a sample from the type Caradoc (Middle Ordovician) Series in Britain, *Geology*, **4**, 505–506.

Thomas, C. & Briden, J. C., 1976. Anomalous geomagnetic field during the Late Ordovician, *Nature*, **259**, 380–382.

Wedd, C. B., Smith, B., King, W. B. R. & Wray, D. A., 1922. The country around Oswestry: explanation of sheet 137, 234 pages, *Mem. geol. surv. England and Wales*, HMSO, London.

Whittard, W. F., 1932. The stratigraphy of the Valentian Rocks of Shropshire, the Longmynd-Shelve and Breidden outcrops, *Quart. J. geol. Soc. London.*, **88**, 859–902.